

Automation Hydroponic Plant Protection from Rain

1. Introduction:

This project aims to design an automatic hydroponic plant cover that will shield the hydroponic plants from rain to maintain the nutrients by preventing it from leaching and overwatering. This report details the project outcomes and performance, design effectiveness, lessons learned and evaluation of the project.

2. Project Outcomes and Performance;

- **Functionality:**

The final circuit successfully functions as intended. When it is raining the rain sensor detects the rainwater, causing the motors to spin and roll open the roof using a belt system. When the roof has opened all the way, the sliding bar hits the limit switch and stops the bar from pulling the roof. When the rain is over and the rain sensor is dry, the motors start spinning in reverse to close the roof until the bar hits the first limit switch where the bar rests at the beginning before the rain, resulting in the roof rolls up perfectly back to where it started.

- **Performance Analysis:**

The performance of the system can be viewed from some aspects:

1. Responsiveness

The system responds promptly to rainfall detection, with the motor activating within approximately 1 second after the rain sensor becoming wet. This quick response helps to prevent plants from being exposed to early drops of rain.

2. Transition Accuracy

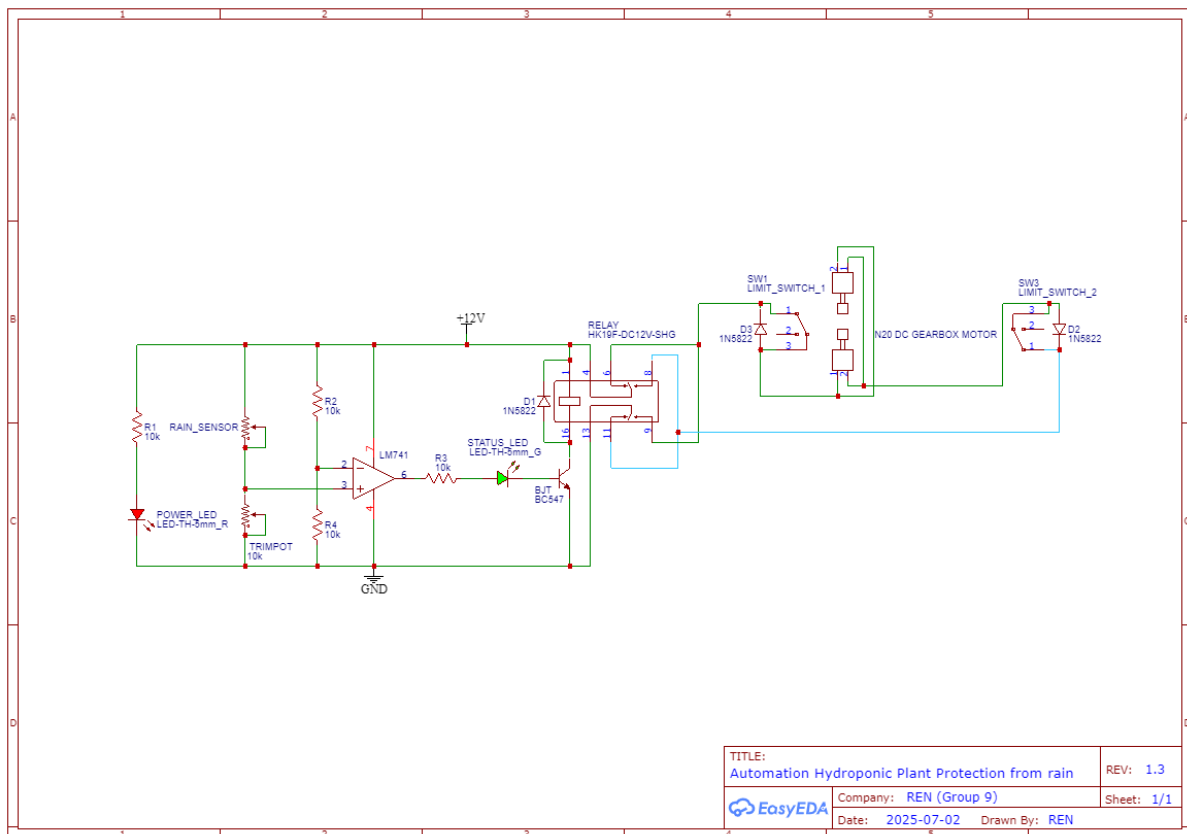
The use of limit switches provides reliable endpoint detection, preventing the tarp from overextending. During testing, the tarp consistently stopped at the desired position both when opening and closing.

3. Mechanical Stability

Mechanically, the system performs reliably. The use of a belt and pulley system allows for smooth tarp movement. No slippage or mechanical failure was observed during repeated tests.

Overall, the system demonstrates strong performance in both electrical and mechanical aspects. Rain detection triggers the tarp unrolling sequence with minimal delay. Limit switches effectively prevent over-extension of the tarp, and the belt mechanism ensures smooth and failure-free operation. No functional or electrical failures were observed, and the system returned to its default position reliably after each cycle.

- **Circuit Diagram (Final Version):**



- **Component Performance:**

All of the components performed as expected. The rain sensor effectively senses rain water. The Op-Amp successfully works as a comparator whether the rain sensor is wet or dry. The BJT successfully controlled the current for switching the pole in the DPDT relay. The DPDT relay successfully controlled the direction of rotation of the motor. The diodes successfully kept the current flow through the limit switch only, making the limit switches effectively functional to stop the current from flowing when the switch is pressed. The limit switches effectively

stops the current from flowing when the rod presses the limit switch, which is when the roof is fully open or fully closed.

- **Power Consumption Analysis:**

The measured power consumption is relatively low, making it suitable for continuous operation using a 12V power supply. Below is the breakdown of the current consumption during each operation cycle:

1. Idle state with sensor dry and tarp fully rolled:
 - ❖ Current: ~ 2 mA
 - ❖ In this state, the op-amp output is off, and the motor is not active. The system only draws minimal current for sensor monitoring and power indicator.
2. Tarp unrolling (rain detected, motor opening the tarp):
 - ❖ Current: ~ 345 mA
 - ❖ This is the highest power consumption phase, as the motor is actively working to unroll the tarp to cover the plants.
3. Fully unrolled state (rain detected, tarp fully opened):
 - ❖ Current: ~ 19 mA
 - ❖ The motor has stopped moving. Current consumption in this state comes mainly from active op-amp circuit and relay holding current.
4. Tarp rolling back (sensor dry, motor retracting the tarp):
 - ❖ Current: ~ 320 mA
 - ❖ The motor is again active, but slightly lower current than during unrolling, due to inactivity of the op-amp circuit and inactivity of relay when pulling the tarp.

Overall, the system remains energy efficient, with high current draw only during motor activation phase (unrolling and rolling the tarp). In idle states, the consumption is minimal, averaging between 2 mA and 19 mA, ensuring that the system does not impose significant load on the power supply when inactive.

3. Design and Implementation Evaluation:

- **Design Effectiveness:**

The system design functions smoothly and effectively fulfills its intended purpose. The automatic cover mechanism, driven by a motor, responds reliably to the rain sensor, opening and closing the tarp without error. The limit switches accurately detect the end positions and prevent overextension. There is still room for improvement—future iterations could benefit from implementing reed (magnetic) switches to reduce mechanical wear and a belt tensioning mechanism to allow tension adjustments without disassembling the system.

- **Implementation Quality:**

The prototype is practical and ready for real-world outdoor application. The structure is water-resistant, and the mechanical-electrical integration works reliably. The entire system can operate immediately in a hydroponic environment. However, the component mounting process still requires improvement. Using proper fixtures such as screws or brackets instead of relying on glue gun or super glue would result in a more durable and professional build.

- **Component Selection Evaluation:**

Component selection was appropriate and well-matched to the system's requirements. Each component—rain sensor, op-amp as comparator, DPDT relay, and limit switches—performed as expected. The decision to reduce the number of motors from four to two was particularly effective, as it simplified the system and resolved previous synchronization issues without compromising power or functionality.

4. Challenges and Lessons Learned:

- Summary of Key Challenges:
 - Creating multiple early prototypes quickly helped identify design flaws early in the process.
 - 3D modeling was a significant and time consuming hurdle, requiring much learning and practice to produce precise and functional parts.
 - Selecting specific materials and components should have been done earlier to avoid delays.
 - Ordering components too close to the deadline created timing issues and delivery concerns.

- Mounting components using super glue and glue guns was suboptimal, especially for parts under mechanical stress, as it not only led to stability concerns but also made repositioning difficult due to the permanent nature of the adhesive.
- The tarp (rain cover material) occasionally caused problems during motor operation. The motor was intended to rotate counterclockwise to unroll the tarp, but after reaching the endpoint—or after repeated use—the tarp sometimes folded incorrectly and forced the motor to reverse its direction (clockwise). This misalignment resulted in the system stopping unexpectedly or failing to operate correctly. This issue occurred because the tarp roller became loose, causing the material to fold in the wrong direction. The problem was resolved by introducing a belt drive system and installing an additional guide pipe above the roller, which ensured the tarp would exit the roller properly in the correct direction.
- **Lessons Learned:**
 - Starting the project earlier provides more time for experimentation and troubleshooting.
 - Mastery of 3D printing techniques (e.g., brim, support structures) greatly improves the quality of printed parts.
 - Maintaining detailed wiring documentation and circuit logic understanding from the start helps avoid assembly errors.
 - Preparing backup components and considering alternatives in advance is crucial in case of failures or wrong selections.

5. Conclusion and Future Work:

- **Overall Project Success:**

The project was successful. The product and the final circuit functions as intended, meeting all the project goals. The automatic roof effectively rolls open when the rain sensor detects rainwater, and closes back when the rain sensor is dry. The performance of the circuit is stable and reliable. We are pleased with the outcome of the project.

- **Future Work:**

For future work, we could add a knob system for adjusting the tension of the belt without having to take off the mountings to adjust it. We also could change the limit switches into reed switches that use magnets to open or close the switch, without worrying the switch will be weary or starting to loosen after getting pressed frequently. Finally, we could make the rod slightly hover by adding more tension to the belt, to prevent the friction of the rod scraping the PVC foam board after it is used for a large amount of time.

Contribution Statement

Focus on Circuit Design, Component Selection and Team Coordination

Student Name: Rafael Marvin Jonathan

Student ID: 2802416835

Contribution Statement:

“My primary contribution was in initiating the project idea and translating it into a clear concept for the team. I explained the overall functionality of the system, including how sensors, relays, and mechanical components would work together to automatically open and close a tarp based on rain detection.

Together with Nicholas, I selected the components used in the circuit. I was then responsible for designing the schematic circuit diagram and creating the PCB layout. I also performed circuit simulations both on a breadboard and using the Falstad simulator to validate the design prior to PCB fabrication.

During the hardware implementation phase, I was assisted by my team member in the PCB engraving process and in soldering components onto the board. Along with Eric and Nicholas, I planned and installed the wiring throughout the system.

I also proposed the design of the mechanical system, which incorporated a timing belt and pulley mechanism to drive the sliding bar that opens and closes the cover.

As the team leader, I coordinated meetings, delegated tasks, managed the document submission process, and took on the role of both project secretary and treasurer by keeping track of progress updates and all project-related expenditures.”

Focus on Mechanical Design, Component Selection & Installation

Student Name: Nicholas Laurensius Halim

Student ID: 2802428072

Contribution Statement:

“My primary contributions to the project focused on the mechanical design and implementation aspects. I was responsible for designing and preparing 3D models for components that were later fabricated using 3D printing, ensuring that parts were

dimensionally accurate and fit well into the system. I also assisted in selecting suitable components for both electrical and mechanical functionality.

Additionally, I contributed significantly to problem-solving during the prototyping phase—such as proposing and implementing the use of an additional guide pipe above the tarp roller to ensure the tarp unrolled correctly. I actively participated in testing sessions alongside other team members and assisted in the physical installation and integration of the project components, ensuring that the system functioned smoothly as a complete unit."

Focus on Integrating Components, Product Finishing & Documentation

Student Name: Eric Christian Darmawan

Student ID: 2802391681

Contribution Statement:

"My primary contribution to the project focused on helping Nicholas and Rafael on the PCB engraving process, building the frame, and integrating the circuit to the frame. I also helped prepare and slice the 3D models made by Nicholas and Rafael before printing on the 3D printers. I also contributed in arranging the cables so the final product has a cleaner look.

Additionally, I contributed significantly to do a documentation of our team's journey from the start of the project until the final product, and edited it for submission requirements. I took all the pictures and videos of the process, from the very first iteration of the product until the final iteration of the product."

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